



Left Ventricular Unloading in Anterior STEMI without Shock: The STEMI Door to Unload (DTU) Randomized Controlled Trial

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PII: S0735-1097(26)05923-1

DOI: <https://doi.org/10.1016/j.jacc.2026.03.071>

Reference: JAC 33125

To appear in: JACC

Received Date: 12 March 2026

Revised Date: 23 March 2026

Accepted Date: 24 March 2026

Please cite this article as: Kapur NK, Mangner N, Aghili N, Faraz H, McElwee S, Nazir R, Chieffo A, Jolly S, Bieniarz M, Cuculi F, George Z, Katopodis J, Manly D, Metzger C, Noel T, Polzin A, Rana G, Rengifo-Moreno P, Westenfeld R, Bossard M, Nordbeck P, Reddy N, Woitek F, Yau R, Facemire C, Parikh A, Gao Y, Awad H, Moretz J, Christanday G, Sood P, Bilazarian S, Simonton CA, Han Y, Marx SO, Burkhoff D, Udelson J, Moses J, Kimmelstiel C, Karas R, Stone GW, O'Neill W, Left Ventricular Unloading in Anterior STEMI without Shock: The STEMI Door to Unload (DTU) Randomized Controlled Trial, *JACC* (2026), doi: <https://doi.org/10.1016/j.jacc.2026.03.071>.

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Brief title: Unloading and Delayed PCI in Anterior STEMI

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Funding: Johnson & Johnson MedTech Heart Recovery

Disclosures

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Mark C. Bieniarz, MD: Consulting fees, honoraria/payments, travel/meeting support (Edward Life Sciences, Abiomed/ J&J Medtech Heart Recovery) and participation in Advisory Board (Abiomed/J&J Medtech Heart Recovery), grants/contracts (Abbot Vascular, Cordis, Boston Scientific, Sanofi SA), royalties/consulting fees (Abbott Vascular, Abiomed/J&J Medtech Heart Recovery, Amgen, AstraZeneca, Bayer, Biosensors SA, Boehringer Ingelheim, Cordis, Daichi, MedAlliance SA, Novartis, Novo Nordisk, Viatrix, Vifor, Sis Medical)

Matthias Bossard, MD: Grants/contracts (Abbott Vascular, Cordis, Boston Scientific, Om Pharma, Sanofi SA), royalties/consulting fees (Abbott Vascular, Abiomed/J&J Medtech Heart Recovery, Amgen, AstraZeneca, Bayer, Biosensors SA, Boehringer Ingelheim, Cordis, Daichi, MedAlliance SA, Novartis, Novo Nordisk, Viatrix, Vifor, Sis Medical, Om Pharma), payment or honoraria for lectures, presentations, speaker bureaus, manuscript writing or educational events (Abiomed/ J&J Medtech Heart Recovery, Boston Scientific, Cordis, Shockwave, SIS Medical, Abbott Vascular, Bayer), support for attending meetings (SIS Medical, Boston Scientific, Biosensors SA,

Om pharma, Abiomed/ J&J Medtech Heart Recovery), DSMB participation or advisory (Amgen AstraZeneca, Bayer, Om Pharma, Novartis, Cordis, Daichii, Novo Nordisk, Boehringer Ingelheim, Sanofi, Boston Scientific)

Daniel Burkhoff, MD: Payment or honoraria for lectures, presentations, speaker bureaus, manuscript writing or educational events (Abiomed/ J&J Medtech Heart Recovery)

Alaide Chieffo, MD: Consulting Fees (Abbot Vascular, Abiomed/ J&J Medtech Heart Recovery, Cordis, Boehringer Ingelheim, Penumbra, Tenamm), honoraria/ payments (Abbot Vascular, Abiomed/ J&J Medtech Heart Recovery, Cordis, Boehringer Ingelheim, Penumbra, Tenamm)

Florim Cuculi, MD: Grants support scientific meetings

Haroon Faraz, MD: Support for present manuscript and attending meetings and/or travel - institutional research contract (Abiomed/ J&J Medtech Heart Recovery)

Zachary George, MD: No compensation for manuscript, royalties, consulting fees nor participation in boards or expert testimony

Yuchi Han, MD: Support for present manuscript and attending meetings and/or travel - study image analysis; grants or contracts with NIH and DOD (not related to study work); payment for expert testimony (Heidell, Pittoni, Murphy & Bach, LLP); leadership fiduciary role (SCMR Board member); NIH EITA Study Section chair

Sanjit Jolly, MD: Consulting fees (Abiomed/ J&J Medtech Heart Recovery, Boston Scientific, Shockwave, Teleflex); honorarias for lectures/presentations (Shockwave, Boston Scientific, SIS Medical)

Richard H. Karas, MD: Manuscript support and consulting fees (Abiomed/J&J Medtech Heart Recovery)

John Katapodis, MD: No compensation for manuscript, royalties, consulting fees nor participation in boards or expert testimony

Carey Kimmelstiel, MD: No compensation for manuscript, royalties, consulting fees nor participation in boards or expert testimony

Norman Mangner, MD: Grant (Abiomed Institutional Grant for Heart Center Dresden); consulting fees (Abbott, Abiomed/J&J Medtech Heart Recovery, Amgen, AstraZeneca, B Braun, Biotronik, Boston Scientific, Cordis, Edwards Lifescience, Inari, Medtronic, Pfizer/BMS, Shockwave); unrestricted educational grants to Heart Center Dresden (Abiomed/J&J Medtech Heart Recovery and Boston Scientific)

David Manly, MD: No compensation for manuscript, royalties, consulting fees nor participation in boards or expert testimony

Steven Marx, MD: No compensation for manuscript, royalties, consulting fees; honoraria received for Clinical Event Committee adjudications (Cardiovascular Research Foundation, Mount Sinai)

Samuel McElwee, MD: No compensation for manuscript, royalties, consulting fees nor participation in boards or expert testimony

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Raja A Nazir, MD: No compensation for manuscript, royalties, consulting fees nor participation in boards or expert testimony

Thomas E Noel, MD: No compensation for manuscript, royalties, consulting fees nor participation in boards or expert testimony

Peter Nordbeck, MD: Grant/consulting fees (Amicus, Abbott, Cardia Dimensions, Sanofi, Takeda, Biotronik, Boston Scientific, Chiesl, Idorsia, Medtronic, Sangamo. Advisory Board or Data safety monitoring: Amicus Therapeutics, Cardia Dimensions, Chiesl, Edwards, Idorsia, Sangamo, Sanofi, Takeda)

William O'Neill, MD: Grants/contracts (Co-PI for STEMI DTU trial, Henry Ford Institution grant support from Abiomed); consultant (Abiomed/J&J Medtech Heart Recovery); travel compensation to major medical meetings for Abiomed/J&J Medtech Heart Recovery and Zoll product discussions; stock ownership and options (Accumed radial and Mirus)

Amin Polzin Prof. Dr. Med.: Consulting fees/payment honoraria (Bayer, BMS, Boehringer, Ingelheim, Pfizer, Sanofi, Abbott, Abiomed/J&J Medtech Heart Recovery, Anylam, AstraZeneca, Corvia Medical, Daiichi Sankyo, Edwards, Medtronic, Occlutech, Phillips)

Gaurav Rana, MD: No compensation for manuscript, royalties, consulting fees; Advisory Board participation (Watchman Advisory Board, Council member compensation)

M. Niranjana Reddy, MD: No compensation for manuscript, royalties, consulting fees nor participation in boards or expert testimony

Pablo Rengifo-Moreno, MD: Consulting fees/payment honoraria (Boston Scientific-Board Advisor, Edwards Life Sciences-Proctor, Therashunt); Advisory Board (Boston Scientific)

Gregg W. Stone, MD: Contracts/grants paid to Mt. Sinai Hospital (Shockwave, Biosense-Webster, Bioventrix, Abbott, Abiomed/J&J Medtech Heart Recovery, Cardiovascular Systems Inc, Phillips, Pulnovo, V-wave, PCORI [via Weill Cornell Medical Center]; consulting fees (Millennia Biopharma, Apollo Therapeutics, Cardiac success, Occlutech, Oxitope, Elixir, Impulse Dynamics, Asceneuron, Myochron, Vesalio, HeartFlow, Colibri, Bioventrix, Abbott, Remote Cardiac Enablement, Valfix, Zoll, Shockwave, Adona Medical, HighLife, Elucid Bio, Aria, Alleviant, FBR Medical, MedHub, Biotyx Medical, Ablative Solutions); payment honoraria (Boehringer Ingelheim); stock or stock options (Cardiac Success, Ancora, Cagent, Applied Therapeutics, Biostar family of funds, SpectraWave, Orchestra Biomed, Aria, Valfix, Xenter, Vascentis)

James Udelson, MD: Manuscript, Committee and Travel support (Steering Committee member, Abiomed/J&J Medtech Heart Recovery)

Felix Woitek, MD: Honoria for lectures/proctoring/consulting (Biotronik, Boston Scientific, Abiomed/J&J Medtech Heart Recovery, Corvia, Insight Lifetech, Teleflex, Shockwave, SMT)

Raymond M Yau, MD: Consulting fees/honoraria/travel (Abiomed/J&J Medtech Heart Recovery, Abbott)

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Acknowledgements

The authors thank the following individuals and organizations for their expertise and contributions to the STEMI DTU trial: STEMI DTU study patients, investigators, and research coordinators; Hao Jiang, PhD (biostatistics); Samantha Polak and Jerry Curran (scientific support); Cardiovascular Research Foundation (ECG and angiographic core labs, DSMB, CEC); Yale Core Lab (echocardiography); Perceptive Imaging (CMR analysis); Duke Core Lab (CMR site qualification and image quality assessment); J&J MedTech Heart Recovery Medical Science Office.

ABSTRACT

Background. Despite rapid percutaneous coronary intervention (PCI) for ST-segment elevation myocardial (STEMI), large infarcts contribute to heart failure and mortality. Left ventricular (LV) wall tension and load are major determinants of infarct size (IS). Preclinical studies identified that delaying reperfusion to permit LV unloading with a trans-valvular micro-axial flow pump (TV-mAFP) reduces IS. We tested whether this combination reduces IS compared with reperfusion alone in patients with anterior STEMI without cardiogenic shock.

Objectives. The STEMI-DTU pivotal trial tested the central hypothesis that the combination of mechanical LV unloading plus a 30-minute delay before PCI reduces infarct size compared with immediate PCI alone in patients with anterior STEMI without cardiogenic shock.

Methods. We conducted an open label, randomized controlled trial at 55 hospitals in the United States, Germany, Italy, the United Kingdom, Switzerland, and Canada. Adults aged 18-85 years with no prior myocardial infarction, presenting with acute anterior STEMI within 1-6 hours of symptom onset before hospital arrival were eligible for inclusion. Patients were randomly assigned (1:1) by study site personnel to either LV unloading with a TV-mAFP for 30 minutes before PCI (treatment group) or PCI alone (control group). The primary outcome was IS normalized to LV mass (IS/LVM) evaluated by cardiac magnetic resonance imaging 3-5 days after PCI and was evaluated in all randomized patients. The trial was registered at clinicaltrials.gov (NCT03947619) and is closed to new participants.

Results. Between December 12, 2019, and September 3, 2024, 527 patients were randomized; 262 patients were assigned to the treatment group and 265 to the control group. Mean age was 61 ± 11 years and 417 (79.1%) were men. Total ischemic time was longer in the treatment arm. IS/LVM was $30.8 \pm 16.2\%$ in the treatment group and $31.9 \pm 16.9\%$ in the control group; mean difference -1.1% [95% CI, -4.2% to 2.0%]; $p=0.50$]. Major bleeding or vascular complications at 30-day follow-up occurred more frequently in the treatment group when compared with either a pre-specified performance goal or the control group.

Conclusions. Combination of a TV-mAFP plus delayed PCI did not reduce infarct size in patients with anterior STEMI without cardiogenic shock compared with PCI alone.

CONDENSED ABSTRACT

The multicenter, randomized STEMI-DTU pivotal trial was designed to test the central hypothesis that the combination of mechanical LV unloading plus a 30-minute period of delay before PCI reduces infarct size compared with immediate PCI alone in patients with anterior STEMI without cardiogenic shock. Combination of LV support with a TV-mAFP plus delayed PCI did not reduce infarct size compared with immediate PCI alone in patients with anterior STEMI without cardiogenic shock. Bleeding events were higher in the treatment arm and may reflect prolonged exposure to high dose peri-procedural anticoagulation driven by the protocolized delay to PCI.

Key words: myocardial infarction, circulatory support, ischemia-reperfusion injury, infarct size

Clinical trial: NCT03947619

INTRODUCTION

ST-segment elevation myocardial infarction (STEMI) is a major cause of morbidity and mortality with >3 million cases occurring annually worldwide (1,2). Coronary occlusion results in decreased myocardial oxygen supply and increased myocardial work and oxygen demand, resulting in a vicious cycle of progressive myocardial damage. The current paradigm for STEMI therapy focuses on rapidly restoring coronary artery blood flow via percutaneous coronary intervention (PCI) to re-establish myocardial oxygen supply. However, despite restoration of epicardial blood flow myocardial perfusion often remains impaired, and restoration of blood flow may induce reperfusion injury, thereby increasing myocardial damage (3). Despite timely reperfusion, every 1% increase in myocardial infarct size is associated with 3.8% and 4.0% increases in one-year all-cause mortality and heart failure (HF) hospitalizations respectively in patients with STEMI, imposing a significant burden on patients and healthcare resources (4,5). For these reasons, new approaches to limit myocardial infarct size in STEMI are needed to address this significant public health problem.

Prior attempts to limit infarct size via vascular conditioning or pharmacologic approaches have not shown clear benefit (6). One critical barrier to these strategies is the mandate for rapid coronary reperfusion in STEMI. Transvalvular microaxial flow pumps (TV-mAFP) reduce ventricular wall stress by displacing blood from the LV into the systemic circulation, thereby decreasing LV pressure and volume. Effective LV unloading directly reduces myocardial oxygen consumption and can attenuate myocardial ischemia (7,8). Multiple preclinical studies have demonstrated that, in the setting of acute MI, LV unloading before coronary reperfusion decreases myocardial ischemia and reduces infarct size compared with reperfusion alone (9,10). Preclinical studies further reported a decrease in infarct size despite delaying reperfusion for up to 60 minutes after initiation of a TV-mAFP compared with immediate reperfusion alone (11-13). Specifically, LV unloading for 30 minutes before

reperfusion reduced myocardial infarct size by nearly 50% compared with immediate reperfusion alone. LV unloading for 15 minutes followed by reperfusion or LV unloading started 30 minutes after reperfusion failed to reduce myocardial infarct size compared to reperfusion alone (13). These studies identified improved myocardial perfusion and activation of cardioprotective molecular signaling pathways that target reperfusion injury as potential mechanisms associated with the observed reduction in infarct size (14).

While unloading the LV prior to reperfusion may reduce infarct size and improve clinical outcomes in STEMI, intentionally delaying reperfusion time in STEMI is not recommended because total ischemic time from symptom onset to balloon angioplasty is directly associated with increased myonecrosis and mortality (15-17). A pilot clinical study confirmed the safety and feasibility of mechanical LV support with a TV-mAFP plus delaying PCI for 30 minutes in patients with anterior STEMI without shock (18). Based on these preclinical and clinical findings, the STEMI-DTU pivotal trial was designed to test the central hypothesis that the combination of mechanical LV unloading plus a 30-minute period of delay before PCI reduces infarct size compared with immediate PCI alone in patients with anterior STEMI without cardiogenic shock.

METHODS

The design of the STEMI Door to Unload (DTU) trial has been reported previously (19). In brief, STEMI DTU was a prospective, multicenter, randomized, controlled, open-label trial to examine whether LV unloading for 30 minutes before PCI reduces infarct size compared with PCI alone without LV unloading, while maintaining an acceptable safety profile in patients with anterior STEMI and without cardiogenic shock. Eligible subjects presenting with anterior STEMI were randomized in a 1:1 ratio to a strategy of primary LV unloading with the Impella CP TV-mAFP (Johnson & Johnson MedTech, Heart Recovery, New Brunswick, NJ) for 30 minutes before PCI versus primary PCI alone with no mechanical LV unloading. The duration of follow-up is 12

months with ongoing observation through 60 months, and the primary endpoint, infarct size expressed as a percentage of LV mass (IS/LVM), was measured by cardiac magnetic resonance (CMR) imaging with late gadolinium enhancement at 3-5 days post-PCI. In addition, non-contrast CMR was performed at 6 months and echocardiograms were performed on days 3-5 and at 90 days. At the time of this report follow-up is complete through 12 months.

The study design was approved by the Food and Drug Administration and registered at clinicaltrials.gov (NCT03947619). An institutional review board or ethics committee approved the trial at each site, and each enrolled patient provided written informed consent. The study was designed by the principal investigators, academic steering committee, and the sponsor. Dr. Kapur is Chief Medical and Scientific Officer of Abiomed (a Johnson & Johnson MedTech company and manufacturer of the device under study). Dr. Kapur served as National Principal Investigator and as a faculty investigator from Tufts University School of Medicine until April 2025, and after the trial had completed enrollment, he transitioned to a role with the sponsor. At that time, Dr. Mangner took over as National Principal Investigator while Dr. Kapur continued involvement with the study from the sponsor side. The academic steering committee members and a Data Safety Monitoring Board had oversight for the safety and feasibility of the trial. A Clinical Events Committee (CEC) that was blinded to treatment allocation independently adjudicated clinical endpoint events in the study. The sponsor was responsible for study conduct, reporting, monitoring, database management and biostatistics, and had access to the raw, de-identified patient-level data. Angiographic, CMR, echocardiography, and ECG images were analyzed by independent core laboratories blinded to the treatment allocation. The sponsor, principal investigators, and steering committee were responsible for study reporting. All data analyses were conducted and validated by the sponsor and were reviewed and verified by the academic steering committee. The sponsor reviewed and approved the manuscript before submission. The steering committee held final responsibility for the decision to submit the manuscript and vouches for the accuracy of the study report and its fidelity to the protocol.

Study population

All patients presenting with anterior STEMI at participating sites were screened for eligibility. The detailed inclusion and exclusion criteria are shown in Supplementary Appendix **Table S1**. Briefly, patients between 18 and 85 years of age with no prior MI, presenting with acute anterior STEMI within 1 to 6 hours of symptom onset before hospital arrival, and who met indications for primary PCI were included. Key exclusion criteria included transfer from an outside hospital where an invasive coronary procedure had already been attempted; cardiogenic shock; unwitnessed cardiac arrest or ≥ 30 minutes of CPR prior to enrollment or any cardiac arrest with impairment in mental status, cognition, or any global or focal neurological deficit; recent administration of fibrinolytic therapy; contraindication to TV-mAFP placement, drug-eluting stents (DES), or contrast CMR; life expectancy ≤ 2 years; or compromised ability to give informed consent or to comply with study procedures, including follow-up CMRs.

Enrollment and randomization

An on-call enrollment committee (EC) member reviewed each patient's presentation against study inclusion/exclusion criteria, including qualifying ECG, to confirm eligibility. All patients provided informed consent before enrollment and prior to initiation of any procedures in the catheterization laboratory. Prior to being enrolled in the study, all consented subjects had femoral angiograms via radial or femoral access approaches to rule out any contraindication for TV-mAFP placement, and LV end-diastolic pressure measurement and left ventriculography to confirm the presence of an anterior MI. The randomization module housed the schema developed by the independent statistician that included the stratification factors, block assignments, and associated treatment arm. Two-step stratification was followed: (1) according to the sum of ST-segment elevation across the precordial leads (V1-V4) (>6 mm or ≤ 6 mm) (prior to protocol v7),

or by site (starting from protocol v7), and (2) according to time from symptom onset to presentation at the study hospital (≤ 3 hours or > 3 hours). Stratified subjects were then placed in blocks of 2, 4 or 6. Subjects were then randomized by study site personnel in a 1:1 ratio to the treatment or control group via a web-based program.

Interventions and procedures

Subjects randomized to the treatment group, TV-mAFP plus delayed PCI, underwent placement of Impella CP through a femoral arterial sheath (22) and the TV-mAFP device was activated once in the LV. Coronary angiography followed by PCI was initiated no earlier than 30 minutes following TV-mAFP placement. Hemodynamic support was required to be maintained at the highest attainable P-level (speed) without causing suction for a minimum of 4 hours. Treatment for systolic hypertension or tachycardia in patients receiving mechanical support was recommended using pharmacological therapies to achieve a systolic blood pressure (SBP) ≤ 130 mmHg and target heart rate ≤ 80 bpm. Operators were permitted to perform bailout PCI at their discretion at any time during the protocolized 30 minutes of TV-mAFP support.

Device and sheath removal were recommended once the following criteria for weaning from TV-mAFP support were met: (1) ≥ 4 hours support achieved at the highest P level; (2) subject tolerates 30 minutes of support at P-3 without hemodynamic instability; (3) partial thromboplastin time or activated clotting time levels demonstrate normalization of anticoagulation; and (4) the facility/health care team is prepared for safe weaning and removal of the device in the catheterization laboratory. If any of these criteria were not met, TV-mAFP support could be extended to a maximum of 24 hours.

Subjects randomized to the control group received PCI alone according to the current standard of care for anterior STEMI patients. Mechanical circulatory support was allowed for patients who developed cardiogenic shock during or after the PCI procedure. Patients in both study groups were treated according to contemporary American and European practice

guidelines for management of STEMI using PCI, both during the index admission and throughout the follow-up period. PCI access site was left to the discretion of the physician and PCI of non-culprit lesions was not permitted during the index procedure. Staged PCI of a non-infarct artery was not permitted prior to the cardiac MRI at 3 to 5 days post-index procedure unless it was necessary to ensure the safety of the subject. The study protocol strongly recommended that all subjects receive guideline-directed medical therapy.

CMR

The CMR protocol to determine infarct size has been described (19). In general, CMR using standard sequences was performed for infarct size, microvascular obstruction (MVO), area at risk (AAR), LV mass, dimensions, and function. Delayed enhancement infarct size imaging was performed after intravenous administration of gadolinium-chelate (0.15-0.20 mmol/kg body weight). Infarct size was expressed as a percentage of total LV mass (IS/LVM) and as a percentage of the AAR. A central core laboratory (Duke Cardiovascular Magnetic Resonance Center, Durham, NC) qualified participating sites and performed quality assessment on the images during the course of the study. Perceptive Imaging (Burlington, MA) provided core lab assessment of CMR parameters on de-identified images without knowledge or access to treatment assignment or clinical outcomes.

Study endpoints

A complete list of study endpoints is detailed in Supplementary Appendix **Table S2**. The primary endpoint was IS/LVM evaluated by CMR at 3-5 days following the index procedure. The key secondary powered effectiveness endpoint was the hierarchical composite of (1) cardiovascular (CV) mortality, (2) cardiogenic shock developing ≥ 24 hours from enrollment, (3) left ventricular assist device or heart transplant, (4) heart failure, (5) implantable cardioverter defibrillator (ICD) or cardiac resynchronization therapy (CRT), and (6) IS/LVM as evaluated by

CMR at 3 to 5 days post-procedure. Components 1-5 of the composite endpoint were evaluated when the last enrolled patient completed 12-month follow-up.

The secondary powered safety endpoint was treatment-related major bleeding or major vascular complications within 30 days, which was evaluated for superiority compared to a pre-defined performance goal of 26.5%, as previously described (19). Additional powered secondary endpoints included infarct size as % of AAR and MVO as % of LV mass as evaluated by CMR at 3 to 5 days post-procedure; left ventricular end-systolic volume (LVESV), end-diastolic volume (LVEDV), and ejection fraction (LVEF) as evaluated by CMR at 6 months post-procedure; and LVESV index (LVESVi) and LVEDV index (LVEDVi) as evaluated by echocardiography at 90 days post-procedure.

Statistical analysis and sample size

A change in sample size from 668 to 516 became necessary after the independent statisticians (Cardiology Research Foundation, CRF) confirmed that the original trial sample size of 668 intention-to-treat (ITT) subjects was overpowered. Upon re-calculation by the sponsor and confirmation by CRF in April 2024, it was determined that 516 ITT subjects would be needed to reach 90% power for analysis of the primary endpoint and key powered secondary effectiveness endpoint. FDA was notified of this sample size change via a formal IDE supplement submission and approved the change.

Assuming an absolute mean difference of 5.0% between treatment and control, which is considered to be the minimum clinically significant difference in infarct size, and a standard deviation of 15.0% in both groups, 382 evaluable subjects were required to provide 90% power to test the hypothesis for the primary endpoint. Based on the assumption of 26% attrition (non-acquired or non-analyzable CMR images), 516 enrolled patients were required to power the primary endpoint. The power analysis for the key powered secondary endpoint and seven additional imaging endpoints have been previously reported (19).

The primary endpoint and powered secondary effectiveness endpoints were analyzed in the ITT population, which included all randomized subjects regardless of their treatment received. Additional analyses of the primary and powered secondary endpoints were performed in the per-protocol (PP) population (Supplementary Appendix **Figure S1**), which was defined as a subset of the ITT population with no major protocol violations and who (a) were randomized to TV-mAFP and received treatment including completion of 30 minutes of ventricular unloading prior to PCI and TV-mAFP support for ≥ 4 hours; or (b) were randomized to the control group and received PCI only. Subjects that the CEC adjudicated as having cardiogenic shock prior to enrollment (based on pre-PCI lactate and hemodynamics) were also excluded from the PP analysis. Safety endpoints were analyzed in the safety population, which included all subjects in whom insertion of a TV-mAFP was attempted during the index procedure.

Baseline characteristics, procedural characteristics, and primary and secondary endpoints were described with summary statistics, where mean, standard deviation, median, quartiles and range were used for continuous variables and frequencies and proportions were used for categorical variables. For the primary endpoint, the null hypothesis was assessed using all evaluable CMR data and the 2-sample t-test at a 2-sided 0.05 level of significance. The assumption for normality was met to justify use of a t-test. The Kolmogorov-Smirnov (K-S) test yielded a non-significant p-value of 0.20, indicating an approximately normal distribution of the data. Due to higher-than-expected attrition of CMR data, the window for CMRs included in the per protocol analysis was expanded from 3-5 days to 2-7 days post-procedure to increase the sample size for evaluation of the primary endpoint. The composite key secondary effectiveness endpoint was evaluated hierarchically using the Finkelstein-Schoenfeld statistic and reported as the win ratio 12 months after randomization of the last enrolled subject.

Twenty-five subgroup analyses were pre-specified from the ITT population (**19**). Three of these pre-specified subgroups were not able to be analyzed due to insufficient data. In addition, based on their clinical presentation, a post hoc subgroup was analyzed according to the

Cardiogenic Shock Working Group–Society of Cardiac Angiography and Interventions (CSWG–SCAI) pre-shock stages at admission (**20**): CSWG–SCAI Stage A patients showed no signs of baseline hypoperfusion (lactate <2 mmol/L) or hypotension (SBP >90 mmHg), while CSWG – SCAI Stage B patients had either isolated hypoperfusion (lactate ≥ 2 mmol/L) or hypotension (SBP ≤ 90 mmHg). A post hoc DanGer SHOCK-like (**21**) subgroup was also analyzed according to patients with both hypotension (SBP <100 mmHg) and hypoperfusion (arterial lactate ≥ 2.5 mmol/L).

RESULTS

A total of 527 patients with anterior STEMI without cardiogenic shock were randomized at 55 sites in the United States, Germany, Italy, the United Kingdom, Switzerland, and Canada from December 12th, 2019 through September 3rd, 2024. Two hundred sixty-two (262) patients were randomized to the treatment group and 265 patients were randomized to the control group (**Figure 1**). Three patients randomized to the treatment group did not receive a TV-mAFP, and no control group patient received a TV-mAFP. CMR was performed in 469 subjects (89.2%). Of these, 443 were evaluable and 412 were performed within 3-5 days. Twelve-month follow-up was available in 459 subjects (96.2%), or 90.5% of all enrolled subjects.

Baseline characteristics were well matched (**Table 1**). The enrolled population had a mean age of 61 ± 11 years and was predominantly male and White/Caucasian. The mean systolic and diastolic blood pressures were 142 ± 25 mmHg and 89 ± 16 mmHg, respectively. Mean heart rate, LV end-diastolic pressure (LVEDP) and sum of ST-segment elevations in the anterior precordial leads were similar between groups.

Symptom onset-to-hospital arrival and hospital arrival-to-first device, either PCI or TV-mAFP in the control and treatment groups respectively, were similar (**Table 2**). No difference was observed in hospital arrival to randomization times between groups. Time to PCI after hospital

arrival was longer in the treatment group by a median of 40 minutes (96 [78, 122] vs. 56 [40, 72] minutes; $p<0.0001$). Total ischemic time, defined as symptom onset-to-PCI, was also longer in the treatment group by a median of 47 minutes (212 [174, 272] vs. 165 [134, 220] minutes; $p<0.0001$; **Table 2, Central Figure**). Median time from TV-mAFP insertion to PCI in the treatment group was 40 (35, 46) minutes and median duration of TV-mAFP support (pre- and post-PCI) was 10.4 (5.5, 18.3) hours. No patient in the treatment group received bailout PCI before completion of the protocolized delay to reperfusion.

Given the need for femoral angiography prior to randomization, femoral vascular access was used in most patients in both groups (**Table 3**). The left anterior descending (LAD) artery was the location of the culprit lesion in 502/513 subjects (97.9%). PCI was performed in 256/262 (97.7%) and 252/265 patients (95.1%) in the treatment and control groups, respectively. Pre-PCI aortic diastolic pressure (86.6 ± 18.0 vs. 77.3 ± 18.5 mmHg; $p<0.0001$) and calculated coronary perfusion pressure (61.4 ± 19.3 vs. 52.5 ± 18.7 mmHg; $p=0.0005$) were higher in the treatment group. The presence of pre-PCI TIMI 0 or 1 flow in the infarct-related artery was higher in the control group (142/254 (55.9%) vs 179/258 (69.4%), $p=0.0038$). Post-PCI TIMI 3 flow was achieved in 210/247 (85.0%) and 205/248 (82.7%) patients in the treatment and control groups, respectively ($p=0.63$). Medication use at discharge was similar between groups (Supplementary Appendix **Table S4**).

The primary endpoint of IS/LVM was $31.4\pm 16.6\%$ in the total study population. Mean IS/LVM was not different between the treatment and control groups in the ITT population ($30.8\pm 16.2\%$ vs. $31.9\pm 16.9\%$; mean difference -1.1% [95% CI, -4.2% to 2.0%], $p=0.50$ (**Table 4, Central Figure**). Results were also similar in the PP population ($31.8\pm 16.2\%$ vs. $33.7\pm 16.1\%$; mean difference -1.9% [95% CI, -5.5% to 1.6%]; $p=0.28$). There was no difference in infarct size between groups when multiple imputation was used to account for missing data ($31.2\pm 16.2\%$ vs. $31.7\pm 16.7\%$; mean difference -0.5% [95% CI, -3.0% to 2.0%]; $p=0.77$). Univariate correlates of IS/LVM were greater total ischemic time, higher baseline LVEDP

and pre-PCI arterial lactate, and lower LVEF and pre-PCI TIMI flow (Supplementary Appendix **Table S5**). The effect of the randomized treatment on IS/LVM was consistent in 23 subgroups (22 pre-specified and 1 post hoc [aortic pulsatility index]) (Supplementary Appendix **Table S6**). Higher baseline arterial lactate levels and CSWG-SCAI Stage B presentation were associated with larger infarcts in the total cohort, but with consistent treatment effects in the randomized groups (Supplementary Appendix **Figure S2**).

The win ratio for the composite secondary effectiveness endpoint in the treatment group compared with the control group in the ITT population was 1.04 (95% CI, 0.84 to 1.28; $p=0.73$; (Supplementary Appendix **Table S7**) indicating no difference between groups. Powered secondary CMR measures at 3-5 days and 6 months and echocardiography measures at 90 days were similar between groups other than the end-systolic volume on CMR at 6 months which was larger in the treatment group ($84.7 \pm 46.3\%$ vs. $75.2 \pm 40.6\%$; mean difference 9.6 [95% CI 0.8 to 18.4]; $p=0.033$; Supplementary Appendix **Table S8**). Non-hierarchical non-powered clinical endpoints at 30 days and 12 months are shown in Supplementary Appendix **Table S9**. No significant differences were observed at either 30 days or 12 months (**Figure 2**). There were no differences between groups in heart failure-related events or other clinical outcomes.

The secondary safety outcome was assessed at 30-day follow-up in 260 treatment group patients in whom TV-mAFP insertion was attempted. Treatment-related major bleeding (BARC types 3-5) or major vascular complications occurred in 80 patients (30.8%; 95% CI, 25.2 to 36.8) (**Table 5**), which exceeded the pre-defined performance goal of 26.5% ($p=0.95$). Treatment-related major bleeding occurred in 79 patients (30.4%) and all were access site-related. Treatment-related major vascular complications occurred in 11 subjects (4.2%) and most were hematomas (**Table 5** and Supplementary Appendix **Table S10**). In a post-hoc analysis, any major bleeding or vascular complications occurred more frequently in the treatment group compared with the control group (89/262 [34.0%] vs. 16/265 [6.0%]; $p<0.01$). Among patients with any major

bleeding in the treatment arm, 88/262 patients (33.6%) experienced any BARC 3 bleeding, of whom 47/88 (53.4%) had a hemoglobin drop of 3-5g/dL; 43/88 (48.9%) had a hemoglobin drop >5g/dL; and zero patients had an intracranial or intraocular hemorrhage (BARC 3c). One patient experienced a fatal bleed (BARC 5) (Supplementary Appendix **Table S10**).

While bleeding and vascular complications were more frequent in the treatment group, among patients with such events 12-month mortality was lower in the treatment group compared with the control group (Supplementary Appendix **Figure S4**). In contrast, mortality rates were similar between groups in patients without bleeding or vascular complications. Major bleeding or major vascular events were significantly associated with 12-month mortality in the control group (31.6% vs. 2.9% in patients without major bleeding or major vascular events, HR 12.74, 95% CI 4.03 to 40.27) but not in the treatment group (7.4% vs. 3.0% respectively, HR 2.46, 95% CI 0.75 to 8.06) ($p_{\text{interaction}}=0.0464$) (Supplementary Appendix **Table S11**).

DISCUSSION

In the multicenter, randomized STEMI-DTU trial, combination of a TV-mAFP plus delayed PCI did not reduce infarct size compared with immediate PCI alone in patients with anterior STEMI without cardiogenic shock. Bleeding events were higher in the treatment arm. The trial evaluated an approach that was mechanistically attractive, but failed to demonstrate benefit with the combined approach.

Total ischemic time is a well-established determinant of infarct size and mortality (**15-17**) and strongly correlated with infarct size in the present study. In the STEMI-DTU trial, a protocolized delay to reperfusion significantly prolonged total ischemic time in the treatment group. The percentage difference between the populations included in the ITT and per-protocol groups was 24.5% and likely reflects the complexity of the protocolized approach to a TV-mAFP first strategy followed by delayed reperfusion. During the period between initiation of the TV-mAFP and PCI, operators were allowed to bail out and perform PCI at their discretion based on

patient stability, yet no patient required PCI during this waiting period, suggesting clinical stability during pre-PCI LV unloading in anterior STEMI. Moreover, despite an average increase in total ischemic time by more than 40 minutes compared with the control group, infarct size was similar between groups. The kinetics of how LV unloading may attenuate the relationship between total ischemic time and infarct size in anterior STEMI without cardiogenic shock requires further investigation. However, based on the results of this trial, there is no rationale for placing a TV-mAFP and intentionally delaying PCI in anterior STEMI without shock.

Consistent with prior studies (22), we observed an inverse association between pre-PCI TIMI flow and infarct size in the overall study population. We also observed a significantly lower incidence of pre-PCI TIMI 0 or 1 flow in the treatment group. Since coronary angiography was not permitted until the end of a 30-minute period after TV-mAFP activation, this observation suggests that use of a TV-mAFP to increase coronary perfusion pressure as observed in the treatment group may increase myocardial blood flow. Consistent with this observation, prior studies have identified that a TV-mAFP increases collateral coronary blood flow and reduces microvascular resistance in STEMI (23-25). Other potential explanations include prolonged exposure to heparin and other anti-platelet agents which are active during this waiting period and may enhance endogenous reperfusion. However, paradoxically an increase in spontaneous reperfusion in the treatment group during the obligate delay period may mask a potential benefit of LV unloading because it would limit the time for activation of cardioprotective pathways. Further studies are required to unravel these important relationships.

Extensive preclinical testing identified that mechanically unloading the LV using a TV-mAFP before reperfusion limits ischemic injury by decreasing wall tension and improving myocardial blood flow (7-9). Moreover, LV unloading activates cardioprotective signaling mechanisms and conditions the myocardium to attenuate reperfusion injury, theoretically further reducing infarct size after PCI. Preclinical studies also identified that the 30 minutes of LV unloading before PCI were necessary and sufficient to reduce infarct size (8-14). Theoretically,

these benefits would also overcome any increase in infarct size necessitated by an obligate delay to reperfusion required for placement of a TV-mAFP. However, in the present adequately powered clinical trial, this combination approach did not reduce nor increase infarct size compared with immediate PCI alone.

Despite the importance of reducing infarct size, no randomized trials have clearly demonstrated benefit of any MCS or pharmacological therapy targeting reperfusion injury. Though not directly tested in this trial, the discrepancy between the prior positive preclinical findings and the current negative clinical results may reflect the fact that TV-mAFP flow is strongly dependent on increased preload and reduced afterload (26,27). Among patients with cardiogenic shock in whom preload is high and afterload is low, pump efficiency will be high (28). In the current population with anterior STEMI without cardiogenic shock, while preload (LVEDP) was high, most patients were hypertensive (high afterload) with a mean blood pressure greater than 140/90. As a result, pump efficiency may be reduced since the device is required to generate flow against a high afterload (26,29). In contrast, much of the preclinical data demonstrating that LV unloading reduces infarct size were obtained under relatively hypotensive conditions more consistent with a pre-cardiogenic shock state.

STEMI-DTU represents one of the largest trials and the most contemporary prospective study of infarct size defined by CMR in patients with anterior STEMI. In addition to total ischemic time and pre-PCI TIMI flow, baseline high LVEDP and pre-PCI arterial lactate levels and low LVEF correlated with increased infarct size, but SBP did not. Since patients with cardiogenic shock, defined as having both hypotension and hypoperfusion, were excluded from the trial, the STEMI-DTU population is enriched with CSWG-SCAI A and B patients. Infarct size was greater among patients with lactate levels above 2.0 or 2.5 mmol/L and among CSWG-SCAI B patients in the total cohort. However, there were no significant interactions between randomized treatment and infarct size according to these parameters. These findings suggest that key metrics can be assessed at baseline during STEMI to identify patients at risk of a large myocardial

infarction and provide new insight into the clinical utility of a pre-PCI lactate level (and LVEDP) as a marker of myocardial damage in pre-shock anterior STEMI patients. These observations may inform future studies in STEMI with and without cardiogenic shock.

Bleeding was a common occurrence in the treatment group of the STEMI-DTU trial and exceeded the established performance goal. The high rate of femoral access in both arms of the study reflects the protocolized need for a femoral angiogram to qualify for randomization and likely influenced overall bleed rates in the study. In addition to the 14 French femoral vascular access sheath required for the TV-mAFP, exposure to high-dose systemic anticoagulation and antiplatelet therapy during the 40-minute delay period in the treatment group, coupled with sustained exposure to anticoagulation for an average of 14 hours until pump removal likely contributed to the high observed bleeding rates. In contrast, the control group received systemic anticoagulation during PCI only. The high rate of femoral access in both groups, driven by the need for a qualifying femoral angiogram, likely also contributed to the high rates of vascular access site bleeding.

In prior STEMI and PCI trials, peri-procedural bleeding is strongly related to subsequent mortality (30). However, in patients who experienced a bleeding or vascular event at 30 days in the present study, one-year mortality was paradoxically lower in the LV unloading group compared with the control group, while mortality was identical between the groups in patients without such events. These findings parallel those reported from the DanGer Shock bleeding sub-study in which BARC 3-5 bleeding was not associated with worsened survival in TV-mAFP - treated patients with cardiogenic shock ($p=0.51$) but was associated in the PCI alone group ($p=0.035$) (31). While these observations do not diminish the importance of minimizing bleeding through best-practice protocols, they provide important clinical context: the bleeding events observed with TV-mAFP therapy may be qualitatively different in their prognostic impact compared with bleeding occurring without hemodynamic support. Moreover, it is well established that access site bleeding is less strongly associated with mortality than is non-access site bleeding

(32). Finally, in the absence of a bleeding or vascular event, one-year mortality rates were low and similar between the treatment and control groups.

While the overall treatment effect was neutral, STEMI DTU was not designed to elucidate offsetting mechanisms that may have affected infarct size within the treatment arm nor was the trial powered for clinical outcomes. The Kaplan-Meier analysis showing the relationship between bleeding and vascular complications and mortality was a post hoc analysis that included relatively low numbers of control arm patients and a low event rate for all-cause mortality. There is a risk of type 1 or type 2 errors in interpreting these findings. Three of the 25 pre-specified subgroups (TV-mAFP P-level, multivessel vs. single vessel CAD, and patients who did vs. did not receive PCI) were not able to be analyzed because there was insufficient data available. Additionally, the study population was approximately 80% male and 20% female, which is consistent with other contemporary STEMI trials. Finally, analyses of secondary endpoints were not adjusted for multiplicity, and given the neutral primary endpoint, all other findings (including in subgroups) should be considered hypothesis generating. Other potential limitations include loss to follow-up, attrition of CMR data, and change in sample size from original SAP to final publication.

In this trial, a strategy of LV support with a TV-mAFP prior to PCI, combined with delayed reperfusion, did not reduce infarct size and was associated with increased bleeding. These findings do not support this approach for reducing infarct size in patients with anterior STEMI without cardiogenic shock. Future studies should build on these findings. As TV-mAFP technologies and training continue to evolve, which may decrease bleeding risk, the novel findings from the STEMI-DTU pivotal trial set the stage for further clinical investigation to reduce infarct size, attenuate bleeding events, and improve the lives of patients with anterior STEMI.

DATA SHARING

Data collected for the study, including individual participant data and a data dictionary defining each field in the set, will be made available 18 months after publication. We will provide deidentified participant data, data dictionary, study protocol, statistical analysis plan, and informed consent form. To gain access to datasets please send an email to cfacemir@its.jnj.com or nkapur5@its.jnj.com. Data will be shared only after approval of the proposal by the principal investigators of the STEMI-DTU trial and will be provided without investigator support and after a signed data agreement has been in place.

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FIGURE LEGENDS

Figure 1. Patient flow in the STEMI DTU pivotal trial.

CMR denotes cardiac magnetic resonance imaging.

Figure 2. All-cause and cardiovascular mortality.

Central Figure. Ischemic time and infarct size in STEMI DTU.

Study design and total ischemic time in patients randomized to the control and treatment groups (top). Absolute difference in infarct size as percent of LV mass (IS/LVM) between the treatment and control groups in the intention-to-treat population (bottom).

Table 1. Baseline characteristics

	Total N=527	Treatment N=262	Control N=265
Age, years	61 ± 11	60 ± 11	62 ± 11
Male	417 (79.1%)	209 (79.8%)	208 (78.5%)
Race			
White/Caucasian	431/484 (89.0%)	209/241 (86.7%)	222/243 (91.4%)
Black/African American	22/484 (4.5%)	17/241 (7.1%)	5/243 (2.1%)
Asian	17/484 (3.5%)	10/241 (4.1%)	7/243 (2.9%)
Other	10/484 (2.1%)	3/241 (1.2%)	7/243 (2.9%)
American Indian/Alaskan Native	2/484 (0.4%)	1/241 (0.4%)	1/243 (0.4%)
Native Hawaiian/Pacific Islander	1/484 (0.2%)	-	1/243 (0.4%)
Multiple	1/484 (0.2%)	1/241 (0.4%)	-
Geography			
United States	364 (69.1%)	179 (68.3%)	185 (69.8%)
Europe	157 (29.8%)	80 (30.5%)	77 (29.1%)
Canada	6 (1.1%)	3 (1.2%)	3 (1.1%)
Body mass index, kg/m ²	28.5 ± 5.2	28.6 ± 5.3	28.3 ± 5.0
Height, cm	173.9 ± 9.3	173.7 ± 9.5	174.1 ± 9.1
Weight, kg	86.2 ± 17.7	86.3 ± 18.0	86.1 ± 17.5
Medical history			
Hypertension (receiving drug therapy)	203/512 (39.7%)	99/254 (39.0%)	104/258 (40.3%)
Current nicotine use	157/504 (31.2%)	73/248 (29.4%)	84/256 (32.8%)
Dyslipidemia (receiving drug therapy)	111/516 (21.5%)	50/257 (19.5%)	61/259 (23.6%)
Diabetes mellitus	111/524 (21.2%)	56/262 (21.4%)	55/262 (21.0%)
COVID-19	59/427 (13.8%)	30/212 (14.2%)	29/215 (13.5%)
Chronic obstructive pulmonary disease	24/516 (4.7%)	12/255 (4.7%)	12/261 (4.6%)
Chronic kidney disease	19/519 (3.7%)	7/257 (2.7%)	12/262 (4.6%)
Peripheral vascular disease	10/516 (1.9%)	4/254 (1.6%)	6/262 (2.3%)
Stroke	7/517 (1.4%)	2/254 (0.8%)	5/263 (1.9%)

At presentation			
Blood pressure, mmHg			
Systolic	142 ± 25	143 ± 25	142 ± 25
Diastolic	89 ± 16	90 ± 16	87 ± 16
Mean arterial pressure, mmHg	106.5 ± 17.8	107.5 ± 17.9	105.5 ± 17.7
Heart rate, core lab, beats/min	81 ± 18	80 ± 18	81 ± 19
Cath lab, pre-percutaneous coronary intervention			
Arterial lactate, mmol/L	2.0 ± 1.2	2.0 ± 1.3	2.1 ± 1.0
Left ventricular end-diastolic pressure, mm Hg	25 ± 9	26 ± 10	24 ± 9
Left ventricular ejection fraction, core lab, %	46 ± 10	46 ± 10	46 ± 11
Anterior ST-segment elevation sum, core lab, mm	9.7 ± 6.4	9.6 ± 6.7	9.7 ± 6.2

Continuous data are mean ± SD. Binary data are n (%).

Table 2. Timing parameters

	Treatment N=262	Control N=265	p-value
Symptom onset-to-hospital arrival	n=262	n=265	
Symptom onset-to-arrival, median (IQR), min	106 (72, 156)	104 (75, 156)	0.86
Arrival-to-randomization	n=261	n=264	
Arrival-to-randomization, median (IQR), min	38 (29, 61)	43 (29, 61)	0.35
Arrival-to-first device (TV-mAFP or PCI)	n=262	n=253	
Arrival-to-first device, median (IQR), min	54 (41, 77)	56 (40, 72)	0.94
Arrival-to-coronary balloon (PCI)	n=255	n=253	
Arrival-to-coronary balloon, median (IQR), min	96 (78, 122)	56 (40, 72)	<0.0001
Symptom onset-to-first device (TV-mAFP or PCI)	n=262	n=253	
Symptom onset-to-first device, median (IQR), min	175 (136, 228)	165 (134, 220)	0.45
Symptom onset-to-coronary balloon (total ischemic time)	n=255	n=253	
Symptom onset-to-coronary balloon, median (IQR), min	212 (174, 272)	165 (134, 220)	<0.0001
TV-mAFP -to-coronary balloon	n=252	-	-
TV-mAFP-to-coronary balloon, median (IQR), min	40 (35, 46)	-	-
Duration of TV-mAFP support	n=259	-	
Duration of TV-mAFP support, median (IQR), hrs	10.4 (5.5, 18.3)	-	-

TV-mAFP, Transvalvular microaxial flow pump; PCI, Primary percutaneous coronary intervention.

Table 3. Procedural characteristics

	Total N=527	Treatment N=262	Control N=265	p-value
Vascular access site				
Radial	96 (18.2%)	53 (20.2%)	43 (16.2%)	0.23
Femoral	429 (81.4%)	207 (79.0%)	222 (83.8%)	0.16
PCI				
Performed	508 (96.4%)	256 (97.7%)	252 (95.1%)	0.11
Thrombectomy	102 (19.4%)	46 (17.6%)	56 (21.1%)	0.30
Not performed	19 (3.6%)	6 (2.3%)	13 (4.9%)	0.11
CABG instead	8 (1.5%)	3 (1.2%)	5 (1.9%)	
Technical limitations	3 (0.6%)	-	3 (1.1%)	
No culprit lesion identified	8 (1.5%)	3 (1.2%)	5 (1.9%)	
Non-culprit lesions treated during the index PCI	42 (8.0%)	21 (8.0%)	21 (7.9%)	0.97
Staged PCI procedure after CMR	49 (9.3%)	19 (7.3%)	30 (11.3%)	0.11
Pre-PCI TV-mAFP pressures (30 min after pump)				
Aortic diastolic pressure, mmHg	81.6 ± 18.9	86.6 ± 18.0	77.3 ± 18.5	<0.0001
Calculated coronary perfusion pressure, mmHg ^a	56.6 ± 19.5	61.4 ± 19.3	52.5 ± 18.7	0.0005
Core laboratory angiographic assessments				
Culprit lesion location ^b				0.88
Left anterior descending	502/513 (97.9%)	249/254 (98.0%)	253/259 (97.7%)	
Proximal	230/513 (44.8%)	114/254 (44.9%)	116/259 (44.8%)	
Mid	266/513 (51.9%)	132/254 (52.0%)	134/259 (51.7%)	
Distal	6/513 (1.2%)	3/254 (1.2%)	3/259 (1.2%)	
First diagonal branch	4/513 (0.8%)	3/254 (1.2%)	1/259 (0.4%)	
Left main coronary artery	1/513 (0.2%)	-	1/259 (0.4%)	
Right coronary artery	4/513 (0.8%)	1/254 (0.4%)	3/259 (1.1%)	

Left circumflex artery	-	-	-	
Other	2/513 (0.4%)	1/254 (0.4%)		
Total lesion length, mm	27.0 ± 13.7	27.5 ± 12.7	26.3 ± 14.9	0.45
Total stent length, mm	32.5 ± 14.8	32.7 ± 14.2	32.3 ± 15.5	0.77
Pre-PCI TIMI flow, culprit vessel				0.0038
0	277/512 (54.1%)	120/254 (47.2%)	157/258 (60.9%)	
1	44/512 (8.6%)	22/254 (8.7%)	22/258 (8.5%)	
2	84/512 (16.4%)	55/254 (21.7%)	29/258 (11.2%)	
3	107/512 (20.9%)	57/254 (22.4%)	50/258 (19.4%)	
Post-PCI TIMI flow, culprit vessel				0.63
0	3/495 (0.6%)	2/247 (0.8%)	1/248 (0.4%)	
1	1/495 (0.2%)	-	1/248 (0.4%)	
2	76/495 (15.4%)	35/247 (14.2%)	41/248 (16.5%)	
3	415/495 (83.8%)	210/247 (85.0%)	205/248 (82.7%)	

^a Calculated as Aortic diastolic pressure - Left ventricle end diastolic pressure.

^b Angiographic images were not assessed in 6 cases (5 Treatment, 1 Control). No culprit lesions were identified in 8 cases (3 Treatment, 5 Control).

TV-mAFP, Transvalvular microaxial flow pump; PCI, Primary percutaneous coronary intervention; CABG, Coronary artery bypass graft; CMR, Cardiac magnetic resonance.

Table 4. Primary endpoint. Infarct size as percentage of LV myocardial mass

Parameter	Total N=527	Treatment N=262	Control N=265	Mean difference (95% CI)	p-value
<u>Intent-to-treat population</u>	n=443	n=220	n=223		
Mean $\pm$ SD, %	31.4 $\pm$ 16.6	30.8 $\pm$ 16.2	31.9 $\pm$ 16.9	-1.1 (-4.2, 2.0)	0.50
Median (IQR), %	31.9 (19.6, 43.4)	31.8 (19.4, 43.5)	31.9 (19.7, 43.4)		0.51
<u>Per-protocol population</u>	n=323	n=153	n=170		
Mean $\pm$ SD, %	32.8 $\pm$ 16.1	31.8 $\pm$ 16.2	33.7 $\pm$ 16.1	-1.9 (-5.5, 1.6)	0.28
Median (IQR), %	34.3 (21.7, 44.8)	32.4 (21.3, 44.1)	36.1 (22.3, 45.4)		0.26

Table 5. Key secondary safety endpoint

Treatment group (Pre-specified safety population) (n=260)	Rate	95% C.I.	Performance goal	P-value
Treatment-related (Device-specific) major bleeding (BARC 3-5) or major vascular complication (30-day)	80 (30.8%)	25.2%, 36.8%	26.5%	0.95
Major bleeding (BARC 3-5)	79 (30.4%)	24.9%, 36.4%		
Total BARC 3	79 (30.4)	24.9%, 36.4%		
BARC 3a	40 (15.4%)	11.2%, 20.4%		
BARC 3b	39 (15.4%)	10.9%, 19.9%		
BARC 3c	0 (0%)	0%, 1.4%		
BARC 4	-	-		
BARC 5	1 (0.4%)	0%, 2.1%		
Treatment-related major vascular complication	11 (4.2%)	2.1%, 7.4%		

BARC 3a: Overt bleeding with HgB drop 3-<5 g/dl; Any transfusion with overt bleeding.

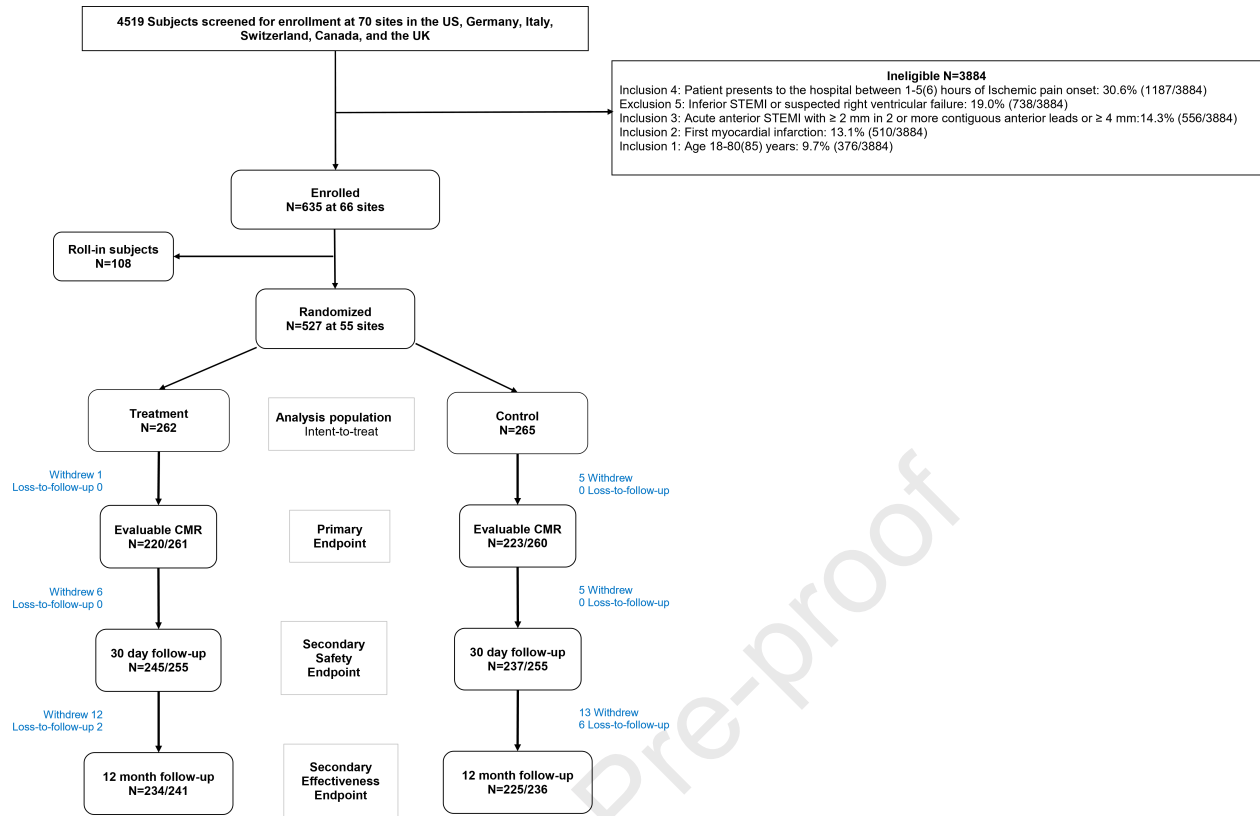
BARC 3b: Overt bleeding with HgB drop ≥5g/dl; Cardiac tamponade; Requiring surgical intervention; Requiring vasoactive agents.

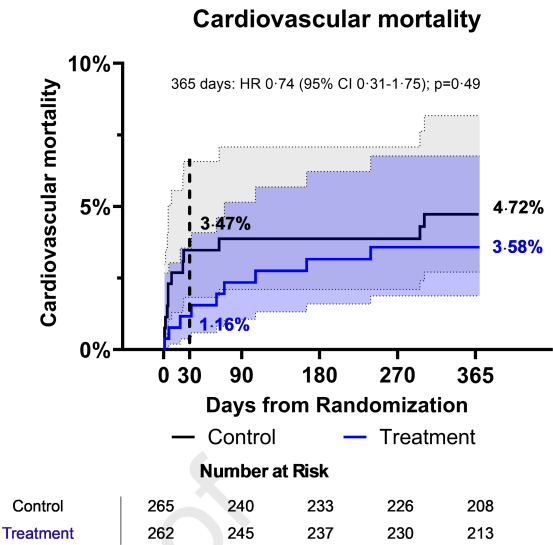
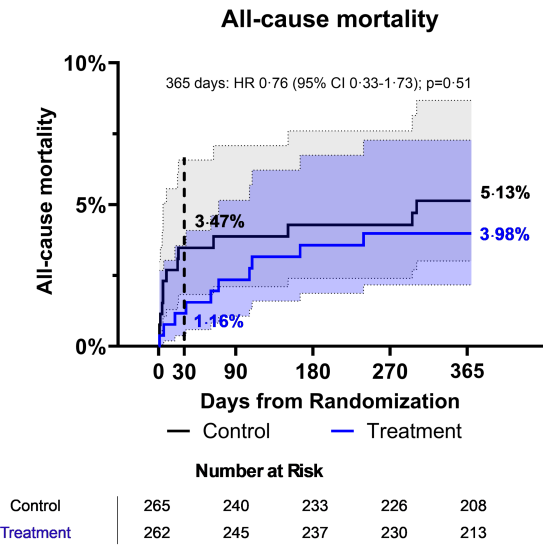
BARC 3c: Intracranial hemorrhage; Intraocular bleed compromising vision.

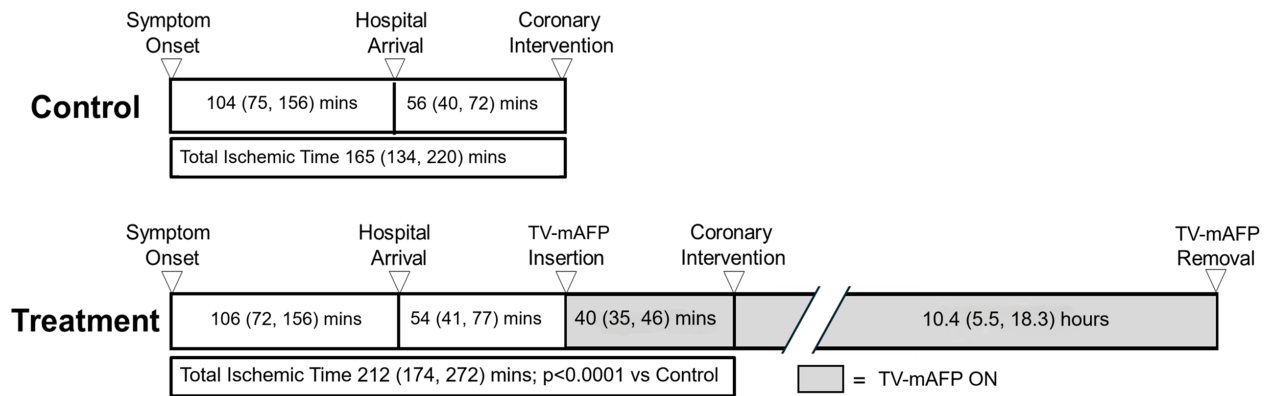
BARC 4: CABG associated bleeding.

BARC 5: Fatal bleeding.

BARC, Bleeding Academic Research Consortium; HgB, Hemoglobin; CABG, Coronary artery bypass graft







Infarct Size / LV Mass (%)

